USN													Course Code:	24CS310

JSS MAHAVIDYAPEETHA

JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

III Semester BE Degree Examination

Department of Computer Science and Engineering

OBJECT ORIENTED PROGRAMMING USING C++

Duration: 3 Hours | Max. Marks: 100

Instructions:

- All questions in PART-A is compulsory.
- Part-B has internal choice.

PART – A									
Q.NO	1	CO	1	CD	L2	PI	1.3.1	MARKS	10
Explain any five features of Object-Oriented Programming (OOP) in detail with suitable examples.									
Q.NO	2	CO	2	CD	L2	PI	1.3.1	MARKS	10
Explain the different types of constructors in C++ with suitable examples for each. Describe how the compiler invokes each type of constructor.									
Q.NO	3	CO	3	CD	L2	PI	1.4.1	MARKS	10
Explain the various types of inheritance in C++. Illustrate each type with a neat diagram and an example program.									
Q.NO	4	CO	4	CD	L2	PI	1.4.1	MARKS	10
With a neat diagram, explain input and output data streams. Demonstrate the unformatted I/O member functions with a C++ program.									
Q.NO	5	CO	5	CD	L2	PI	1.1.1	MARKS	10
What is a template? Explain two types of templates supported by C++ with suitable examples for each.									

PART – B									
Q.NO	6	CO	1	CD	L2	PI	1.3.1	MARKS	5+5
a) Write a C++ program to rotate the values of three variables using reference variables such that: x takes the value of y, y takes the value of z and z takes the value of x. b) Explain inline function in C++ with an example program and show the output.									
OR									

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Q.NO	7	CO	1	CD	L2	PI	1.3.1	MARKS	5+5

a) What are default arguments? Write a C++ program that uses a class with a member function that demonstrates default arguments with sample output.
b) Explain dynamic memory allocation and deallocation for an integer with an example program.

Q.NO	8	CO	2	CD	L3	PI	1.1.1	MARKS	5+5

a) How does operator overloading help in converting basic type to class type and class type to basic type? Explain with a sample code.
b) Write a C++ program to create a class Student with data members - id and marks. Overload the == operator using a member function to compare if two students have the same marks.

OR

Q.NO	9	CO	2	CD	L3	PI	1.1.1	MARKS	5+5

a) Describe the structure of a class in C++. Explain how member functions can be defined both inside and outside a class with a sample code.
b) Write a C++ program to create two classes. Define a friend function to access the private members of both classes and display their sum.

Q.NO	10	CO	3	CD	L3	PI	1.4.1	MARKS	10

Write a C++ program to demonstrate a virtual function. Create a base class Shape with a virtual display () function. Derive classes Rectangle and Circle from it, each providing its own display () function to show a suitable message. Use a base class pointer to call the display () function and demonstrate runtime polymorphism.

OR

Q.NO	11	CO	3	CD	L3	PI	1.4.1	MARKS	5+5

a) Explain function overriding in C++ with an example. How does it differ from function overloading?
b) Explain the order of execution of constructors when class C inherits from class A and virtual class B in public mode and an object of class C is created. Illustrate your answer with a C++ program and show the output.

Q.NO	12	CO	4	CD	L2	PI	1.3.1	MARKS	5+5

a) Write a C++ program to create a file named "ITEM", store item name and cost in it, and then read and display the stored data. Use constructor functions to open the file.
b) Explain the following ios format functions with sample code and output: width(), precision(), fill(), setf(),unsetf().

OR

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Q.NO	13	CO	4	CD	L2	PI	1.3.1	MARKS	5+5

a) Explain exception handling mechanism with an example program.
b) Explain user-defined manipulators in C++ with syntax and an example program. Also write the output of the program.

Q.NO	14	CO	5	CD	L3	PI	1.1.1	MARKS	10

Write a C++ program to create a class template Queue. Implement insert_front and delete_rear and display as function templates inside the class. Use this class template to implement a linear queue of integers and a linear queue of characters. Demonstrate the queue operations by displaying the contents.

OR

Q.NO	15	CO	5	CD	L3	PI	1.1.1	MARKS	10

Write a C++ program to create a class template Stack with push, pop and display member functions. Use this template to implement a stack of integers and a stack of characters. Demonstrate the stack operations by displaying the contents.

Course Outcome (CO):	
CO–1	Understand object-oriented programming concepts and be able to design solutions for a given problem.
CO–2	Able to reuse code with extensible Class types, user defined operators and function overloading.
CO–3	Achieve code reusability and extensibility by means of Inheritance and Polymorphism.
CO–4	Incorporate file-handling functions and exceptions for a given application.
CO–5	Design C++ packages with templates for providing solutions to complex problems.

PI	
1.1.1	Apply the knowledge of discrete structures, linear algebra, statistics and numerical techniques to solve problems
1.3.1	Apply engineering fundamentals
1.4.1	Apply theory and principles of computer science and engineering to solve an engineering problem

Cognitive Domain (CD):	
L1	Remember
L2	Understand
L3	Apply
L4	Analyse
L5	Evaluate
L6	Create

--- End ---